



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 8

Team No.: 13

Project Title: Kernel Monitor

Team Members

Roll Number	Student Name	Signature
2520030531	D. Venya Sri	
2520030527	K. Vansika Reddy	
2520030212	N. Rishika Chowdary	
2520030060	J. Hansika	

1. Abstract

The Linux Kernel Monitor is a system monitoring application developed to provide users with a simple and centralized view of important Linux operating system information. In a Linux environment, details such as CPU usage, memory utilization, running processes, system uptime, and load information are available through different commands and system files, making it inconvenient to monitor them separately. This project solves this problem by developing a user-friendly monitoring tool using the C programming language that collects and displays essential system information in one place. The application uses the Linux /proc virtual filesystem and system interfaces to obtain real-time information about CPU performance, memory usage, system uptime, load average, and active processes. It can display process IDs (PIDs), process states, CPU information, total memory, available memory, and memory utilization percentage. A live monitoring feature can also be implemented to automatically refresh the information at regular intervals, allowing users to observe changes in system resource usage. The project demonstrates important Operating System concepts, including process management, memory management, CPU utilization, file handling, system information retrieval, and interaction between user-space applications and kernel-provided information. It can help users understand the current state of a Linux system and identify processes or resources that are consuming significant system resources. Overall, the Linux Kernel Monitor provides a practical and educational way to understand how information maintained by the Linux operating system is accessed, processed, and presented to users through a simple monitoring application.

2. Problem Statement

Linux operating systems maintain a large amount of information about system resources and running processes, such as CPU utilization, memory usage, process IDs, process states, system uptime, and system load. However, this information is generally accessed through different commands and system interfaces, making it difficult for users, especially beginners, to monitor the overall system status from a single place. Therefore, there is a need for a simple and centralized Kernel Monitor that can collect and display important system information in an easy-to-understand format. The proposed project addresses this problem by developing a Linux-based monitoring application using C programming and system-level interfaces, particularly the /proc virtual filesystem. The project is relevant to Operating Systems and Systems Programming because it demonstrates how user-space programs interact with information provided by the Linux kernel and how system resources and processes are monitored.

It provides practical understanding of concepts such as process management, CPU utilization, memory management, file handling, system information retrieval, and kernel-user space interaction.

3. Objectives

1. To develop a Linux-based Kernel Monitor that displays important system information in a simple and user-friendly format.
2. To monitor system resources such as CPU usage, memory utilization, system uptime, and load average.
3. To monitor running processes by displaying details such as Process ID (PID), process name, and process state.
4. To understand Operating System concepts such as process management, memory management, file handling, and interaction between user-space applications and the Linux kernel.

4. Proposed Methodology

1. System Information Collection – Collect CPU, memory, uptime, load average, and process information from Linux system interfaces such as the /proc virtual filesystem.
2. Data Processing – Read and process the collected system information using C programming and calculate values such as CPU and memory utilization.
3. Process Monitoring – Identify and display currently running processes along with their Process ID (PID), process name, and process state.
4. Real-Time Monitoring – Refresh the collected information at regular intervals to provide an updated view of system resource usage.
5. User Interface – Present the monitored information through a simple command-line dashboard/menu, allowing users to select and view different system statistics.
6. Testing and Validation – Test the application under different system loads and compare the displayed results with standard Linux monitoring commands to verify accuracy.

5. Operating Systems Concepts / Linux APIs Used

(List the relevant Operating Systems concepts, Linux system calls, POSIX APIs, or mechanisms that will be used.)

OS Concept / Linux API / System Call	Purpose in the Project
User Space and Kernel Space	The Kernel Monitor runs in user space and obtains system information provided by the Linux kernel.
/proc Virtual Filesystem	Used to collect CPU, memory, uptime, load average, and process information.
open() System Call	Used to open /proc files for reading system and process information.
read() System Call	Used to read the required information from /proc files.
close() System Call	Used to close files after reading the required information.
Process Management	Used to identify and display running processes, PIDs, process names, and process states.
Memory Management	Used to monitor total, available, and used memory of the Linux system.

CPU Utilization	Used to calculate and display the current CPU usage of the system.
File Descriptors	Used for accessing /proc files through Linux file I/O operations.
POSIX Threads (pthread)	Can be used for continuous/live monitoring and updating system statistics.

6. Individual Contribution

(Clearly specify the responsibility of each team member in the project.)

Roll Number	Student Name	Individual Responsibility
2520030531	D. Venya Sri	CPU & System Monitoring – Implemented CPU usage, CPU information, kernel information, system uptime, and load average monitoring.
2520030527	K. Vansika Reddy	Memory Monitoring – Implemented total, used, free, and available memory monitoring and memory usage calculation.
2520030212	N. Rishika Chowdary	Process Monitoring – Implemented detection and display of running processes, including PID, process name, and process state.
2520030060	J. Hansika	Integration & Testing – Combined all modules, developed the user interface/menu and live monitoring, performed testing, debugging, and documentation.

7. Tools / Platforms / Software Used

(List the programming languages, operating system, libraries, tools, and software planned for the project.)

Tool / Platform / Software	Purpose
Linux / Ubuntu	Platform for developing and running the project
C / C++	Programming language used for implementation
Linux System Calls / APIs	Used for system-level operations
Linux File I/O	Used to access and read system information
POSIX APIs	Used for Linux system programming

8. Expected Outcome

The proposed Linux Kernel Monitor is expected to provide a simple and user-friendly application for monitoring important Linux system information. The system will display CPU utilization, memory usage, system uptime, load average, and information about running processes, including process IDs and process states. The application will collect and present this information in a centralized interface and, where implemented, update the displayed statistics periodically for monitoring changes in system resource usage. The project is expected to provide practical understanding of Operating System concepts and Linux system programming, particularly process management, memory management, file I/O, system calls, and interaction between user-space applications and the operating system. This aligns with the form's requirement to describe the expected functionality, output, and outcome of the project.

9. GitHub Repository

GitHub Repository Link: [janumpallyhansika/Kernal-Krew-_OSSP-](https://github.com/janumpallyhansika/Kernal-Krew-_OSSP-)

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____